



CSE6748 Applied Analytics Practicum

Stacking Sats: Dynamic DCA Bitcoin Accumulation

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1. Introduction

Bitcoin is the first and largest cryptocurrency, with a market capitalization exceeding USD 1.47 trillion and over 600,000 daily transactions. Bitcoin pricing has fluctuated rapidly since its first public price of US\$0.003. Its price increased by over 42 million times to a peak of USD 126,210 in 2025, followed by a 52% decline. Despite not being legal tender in most of the world, its more established status and high volatility have attracted high participation from Bitcoin miners and traders keen to profit from price movements.

One of the most popular methods to invest in a tradable asset is dollar-cost averaging (DCA), whereby an investor buys a certain amount of shares or other assets regularly to get lower average purchase prices and reduce risk. The simplest is Uniform DCA, which allocates a fixed dollar amount at regular intervals. The other possibility is Dynamic DCA, which adjusts allocation based on defined rules to achieve higher acquisition efficiency than Uniform DCA.

The goal of this project is to develop and evaluate a daily Dynamic DCA strategy to buy Bitcoin from 2018-01-01 to 2025-12-31 (Figure 1). The project integrated information from various sources including Bitcoin on-chain data, Polymarket prediction markets, Federal Reserve macro data, and Binance exchange datasets. Subsequently, the dataset was fed into a predictive machine learning model to estimate future returns and risk. Finally, a strategy integrating valuation, momentum, risk, and market sentiment signals to improve was developed and backtested.

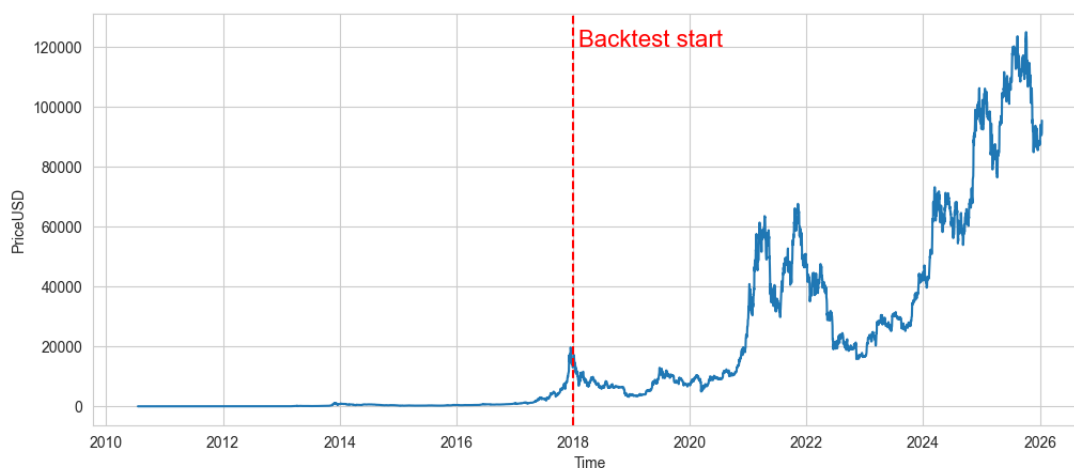


Figure 1. Bitcoin price history

There are three main notebooks which are utilized in this project (<https://github.com/timothyhx1/GT-MSA-Spring-2026-Team-23/>):

- [eda/Part1 Pre-process.ipynb](#): data extraction and pre-processing
- [eda/Part2 Modelling.ipynb](#): dataset analysis and ML training
- [final/Part3 Strategy Optimization.ipynb](#): strategy development, model optimization and backtesting results.

2. Data Import and Feature Engineering

The data sources used in the project are listed in Table 1. Initial on-chain metrics from Coinmetrics and prediction market data from Polymarket were provided. To further enhance information availability, additional trading data from Binance and macro data from Federal Reserve was also imported. The goal of this section is to conduct extensive feature engineering to capture major dimensions of the Bitcoin market: valuation, momentum, risk, and market sentiment. In the next analysis, features with low usefulness or redundant and will be discarded.

Table 1. Data sources for Dynamic DCA strategy development

Type	Source	Data availability
On-chain	Coinmetrics [1]	Since 2010
Technical	Processed from Coinmetrics	Since 2010
Prediction market	Polymarket [2]	Varies. Detailed trade data only since 2025-04-07
Klines daily trade	Binance [3]	Since 2017
Macro	Federal Reserve [4]	Since 2006

2.1. On-chain

Additional features were derived from existing columns to cover key aspects of the Bitcoin market including valuation, momentum, miner stress, exchange flow, volatility, and market activity (Table 2).

Table 2. Additional features created from on-chain dataset

Name	Description	Category
oc_MVRVCur_Z	MVRV Z-score on a rolling 365-day period	Valuation
oc_MVRVCur_grad	MVRV Z-score smoothed gradient (first derivative)	Momentum
oc_MVRVCur_accel	MVRV Z-score smoothed acceleration (second derivative)	Momentum
oc_MVRV_high	1 if CapMrktCurUSD Z-score > 2	Valuation regime
oc_Puell_proxy	daily issuance value divided by 365-day rolling mean. Lower is better buying opportunity.	Miner stress
oc_miner_stress	HashRate divided by 365-day rolling mean	Miner stress
oc_netexchangenorm	(FlowInExUSD - FlowOutExUSD) / CapMrktCurUSD	Exchange flow
oc_flow_ratio	(FlowInExUSD - FlowOutExUSD) / (FlowInExUSD + FlowOutExUSD)	Exchange flow
oc_MVRVCur_vol_Z	MVRV 30-day standard deviation Z-score	Volatility
oc_vol_30d	Rolling standard deviation of log returns	Volatility
oc_vol_30d_Z	Z-score of oc_vol_30d	Volatility
oc_turnover	volume_reported_spot_usd_1d/CapMrktCurUSD	Market activity

2.2. Technical Indicators

Technical indicators used were derived from price data (Table 3).

Table 3. Technical indicators created from PriceUSD data

Name	Description	Category
sma_200	200-day price rolling average	Valuation
smastd_200	200-day price rolling standard deviation	Volatility
distance_sma_200	Price normalized to 200-day moving average	Valuation
ema_200	200-day price EWMA	Valuation
emastd_200	200-day price exponential rolling standard deviation	Volatility
distance_ema_200	Price normalized to 200-day EWMA	Valuation
drawdown	Percentage drawdown compared to all-time-high	Valuation
RSI	Relative strength index indicating oversold or overbought	Valuation
RSI_zscore	Z-score for RSI	Valuation

2.3. Macro

The FRED (Federal Reserve Economic Data) data was added to integrate macroeconomic regime information into the model.

Table 4. FRED (Federal Reserve Economic Data)

Name	Description	Category
M2SL	Money supply to indicate liquidity cycles	Macro regime
WALCL	Fed balance sheet	Macro regime
fed_balance_growth	YoY increase of WALCL	Macro regime
real_rate	DFII10 (10-Year Treasury Inflation-Indexed Constant Maturity Securities)	Macro regime
usd_strength	DTWEXBGS (U.S. Dollar strength)	Macro regime
10y_yield	DGS10 (10-Year Treasury Constant Maturity Securities)	Macro regime
inflation_yoy	CPIAUCSL (Consumer price index)	Macro regime

2.4. Binance

Klines dataset describing daily trading activities and Bitcoin futures data were also included.

Table 5. Binance Klines and futures data

Name	Description	Category
time, open, high, low, close, volume, close_time, quote_volume, trades, taker_buy_base, taker_buy_quote, ignore	Binance Bitcoin exchange Klines dataset	Exchange

funding_rate	Funding rate of Bitcoin futures	Future exchange
markPrice	Mark price of Bitcoin futures	Future exchange
bin_volatility_30d	Price rolling standard deviation	Volatility
bin_volume_zscore	Trading volume Z-score	Volatility

2.5. Polymarket

The Polymarket predictive markets dataset received the highest amount of attention and effort of feature engineering. The dataset included 6 parquet files including “markets”, “tokens”, “trades”, “odds_history”, “event_stats”, and “summary”.

2.5.1 Markets and Tokens

This study focuses on Bitcoin related markets and hence targeted only markets with “Bitcoin” or “BTC” in the titles. A LinearSVC model with a TF-IDF vectorization pipeline was trained to categorize markets with missing categories into one of existing ones (“Crypto”, “Business”, “Global Politics”, “US-current-affairs”). Second, “Crypto” markets related to Bitcoin specifically were recategorized into a new category “Bitcoin”, which is 26.1% of the markets or 20,408 out of 78,321 markets (Figure 2 left).

The “question” column can be categorized as bullish (positive for Bitcoin), bearish (negative for Bitcoin), or neutral depending on the wording or keywords used. For each question, there can be multiple tokens/answers which can be classified as positive or negative sentiment. A trade can be classified bullish, bearish, or neutral based on combined interaction between question framing, token, and transaction direction (Figure 2 right).

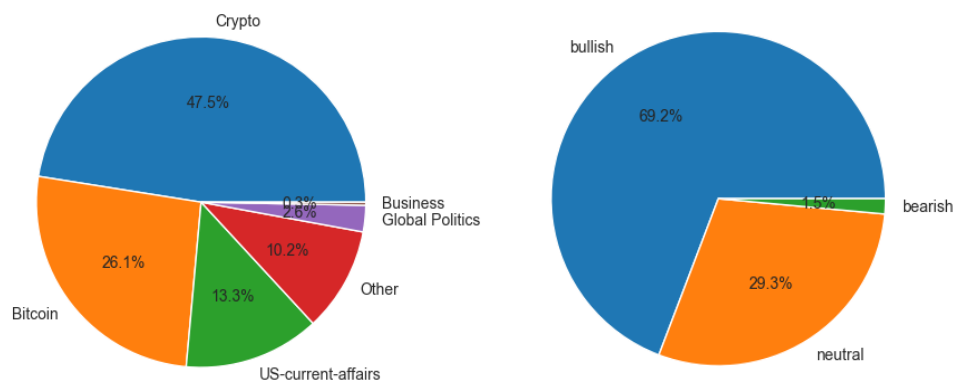


Figure 2. Distribution of Polymarket market categories (left) and Bitcoin market question framing distribution (right)

Markets dataset (20k) → Tokens dataset → Trades dataset (12.4M)

Question e.g.	Question sentiment	Token	Token sentiment	“BUY” Trade	“SELL” Trade
"Bitcoin all time high in Q1?"	Bullish	YES	Positive	Bullish (1)	Bearish (-1)
		NO	Negative	Bearish (-1)	Bullish (1)
"BTC dips below 65k before August?"	Bearish	YES	Positive	Bearish (-1)	Bullish (1)
		NO	Negative	Bullish (1)	Bearish (-1)
"MOG vs. BITCOIN - Most Liked Post Today"	Neutral	MOG	Neutral	Neutral	Neutral
		BITCOIN	Neutral	Neutral	Neutral

Figure 3. Classification of polymarket individual trades based on question framing, token, and buy/sell transaction

2.5.2 Trades

The “trades” dataset was merged with “tokens” to categorize individual trades into bullish, bearish, or neutral. Afterwards, I was able to perform daily aggregates of the trades dataset to generate additional features listed in Table 6.

Table 6. Features derived from Polymarket data

Name	Description
poly_dtrade_count	daily no of trades
poly_dtrade_active_traders	daily no of traders
poly_dtrade_avg_trx	daily average trade value
poly_dtrade_trx	daily total trade value
poly_dtrade_trx_dir	daily net trade value between bullish and bearish transaction
poly_dtrade_trx_std	daily bullish and bearish trade value standard deviation
odds_markets	number of markets with transactions on that day
odds_avg (0 to 1)	average adjusted token price transacted that day (1 being very bullish, 0 being fully bearish)
odds_dispersion	token price standard deviation
odds_breadth	percentage of markets being bullish on that day
odds_long_market	number of markets with end time more than 30 days out. Assumed to be more speculative
odds_long_avg	odds_avg for long markets
odds_short_market	number of markets with end time more 30 days or less. Odds assumed to be more certain
odds_short_avg	odds_avg for short markets
poly_dtrade_trx_zscore / poly_dtrade_trx_dir_zscore / odds_avg_zscore / odds_long_zscore / odds_short_zscore	Z-score for different metrics
poly_dtrade_trx_dir_ratio	poly_dtrade_trx_dir / poly_dtrade_trx or trade imbalance

2.6. Target features

Various target metrics were generated focusing on returns and drawdown risk. Below are the 3 used in the modelling:

- `target_up_60d`: This metric is intended to capture the efficiency of buying Bitcoin today compared to sometime in the next 60 days. A categorical target was selected instead of a continuous one to reduce the impact of noisy returns.

$$target_up_60d = \begin{cases} 1, & ROI_60d_avg > 0 \\ 0, & Otherwise \end{cases}$$

$$ROI_60d_avg = \frac{\overline{Price_{t+1 \text{ to } t+60}}}{Price_t} - 1$$

- `ROI_30d_top30`: Another measure of profitability. With this metric, the ROI threshold changes with respect to whether Bitcoin has had more bull or bear periods up to the training period. Over 2010-2026, the 70th percentile `ROI_30d` is ~15.8%, which has high threshold of profitability. In comparison, `target_up_60d` which uses threshold of 0% and may introduce higher label noise.

$$ROI_30d_top30_t = \begin{cases} 1, & ROI_30d > 70th \text{ percentile of historical } ROI_30d \text{ until time } t \\ 0, & Otherwise \end{cases}$$

$$ROI_30d = \frac{Price_{t+30}}{Price_t} - 1$$

- `Future_60d_min_rel`: Captures downside risk and whether there is a better window to buy in the next 60 days.

$$Future_60d_min_rel = \ln\left(\frac{\text{Average of lowest 5 prices in the next 60 days}}{Price_t}\right)$$

2.7. Culling and cleanup

After all the feature creation, a total of 123 columns were present in the dataset. Most of the uninformative, redundant, or highly correlated features were removed, resulting in a dataset containing 5,659 rows and 86 columns (2 indices, 64 features, and 20 forward looking columns).

3. Modeling

The following section evaluates whether predictive signals exist and whether they remain robust under realistic deployment conditions. A few machine learning algorithms were trained on 64 feature columns to predict the 3 target columns (`target_up_60d`, `ROI_30d_top30`, and `future_60d_min_rel`).

3.1. Simple train-test split

The section compares four algorithms using a simple train and test dataset split:

1. Dataset was split into 80% training set and 20% test set. The test set also excluded 60-day purge period to avoid forward leakage from the training set forward looking target metrics.
2. A 5-fold GridSearchCV was performed on training dataset to optimize hyperparameters for a few sampled supervised machine learning algorithms (XGBoost [5], Logistic Regression, Random Forest, and LightGBM [6]). The GridSearchCV module utilized TimeSeriesSplit with walk-forward validation to avoid forward data leakage by mixing training data from different time periods.
3. As the binary classification outputs were highly noisy with less than 60% accuracy, the prediction probabilities were plotted against forward returns (ROI_60d_avg) instead.

Of the 4 models, XGBoost demonstrated the strongest correlation and monotonic relationship to probability quintiles (Figure 4). Logistic regression was unable to capture the more complex non-linear relationships in the data. RandomForest and LightGBM had decent performance but XGBoost predicted probabilities showed significantly larger range and higher correlation which is more suitable for strategy integration.

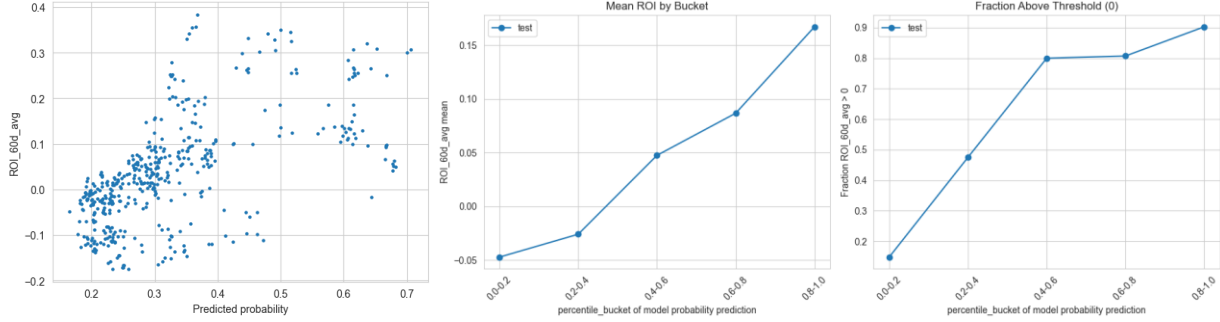


Figure 4. ROI_60d_avg compared to XGBoost test set prediction probability on target_up_60d (left) [corr: 0.66, p-val: 9e-58]. ROI_60d_avg mean (middle) and fraction ROI_60d_avg > 0 (right) compared to prediction probability percentile bucket.

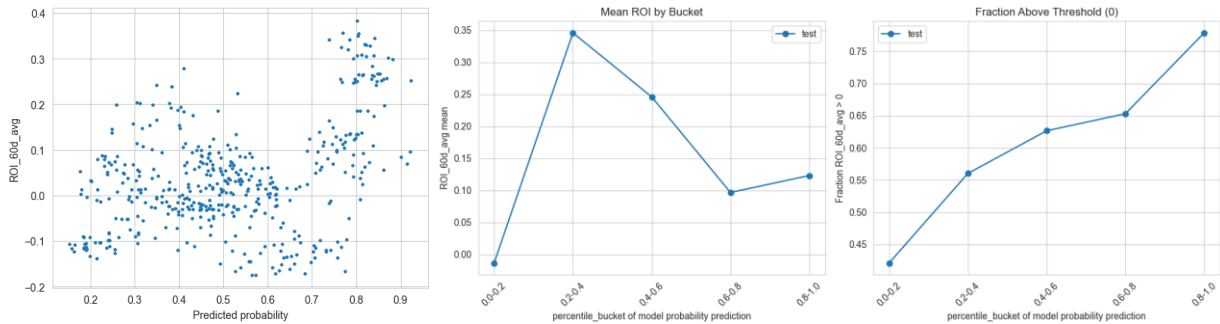


Figure 5. ROI_60d_avg compared to Logistic Regression test set prediction probability on target_up_60d (left) [corr: 0.32, p-val: 6e-12]. ROI_60d_avg mean (middle) and fraction ROI_60d_avg > 0 (right) compared to prediction probability percentile bucket.

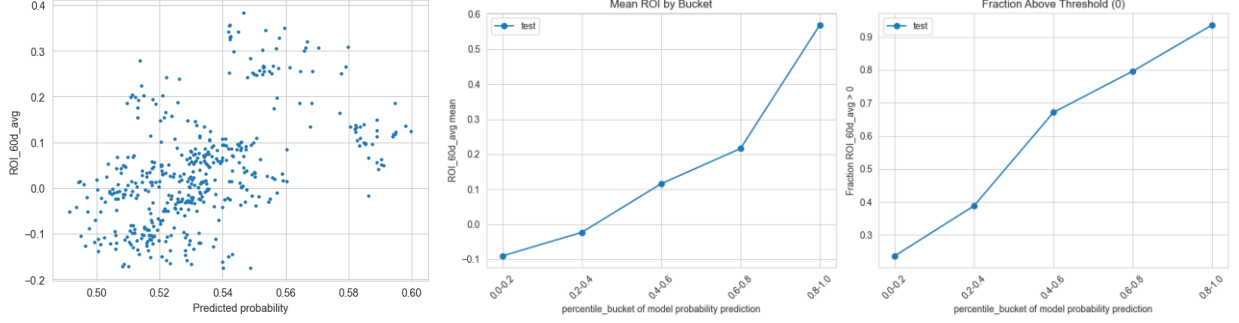


Figure 6. ROI_60d_avg compared to Random Forest test set prediction probability on target_up_60d (left) [corr: 0.53, p-val: $5e-34$]. ROI_60d_avg mean (middle) and fraction ROI_60d_avg > 0 (right) compared to prediction probability percentile bucket.

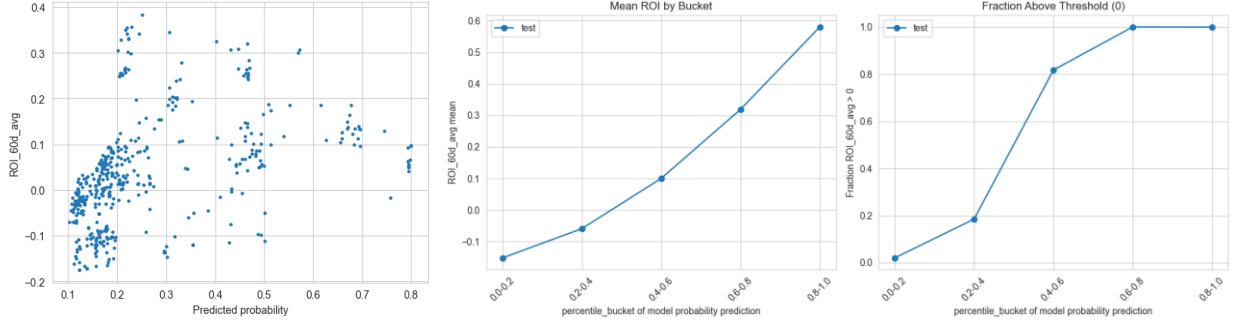


Figure 7. ROI_60d_avg compared to LightGBM test set prediction probability on target_up_60d (left) [corr: 0.62, p-val: $1e-48$]. ROI_60d_avg mean (middle) and fraction ROI_60d_avg > 0 (right) compared to prediction probability percentile bucket.

Similar application of XGBoost was used for the other two target metrics (), exhibiting decent performance although not as strong as that for target_up_60d.

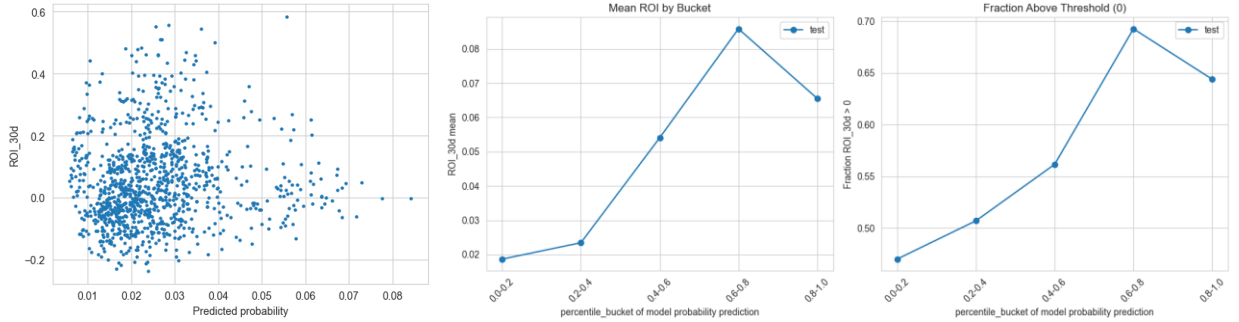


Figure 8. ROI_30d compared to XGBoost test set prediction probability on ROI_30d_top30 (left) [corr: 0.15, p-val: $1.5e-6$]. ROI_60d_avg mean (middle) and fraction ROI_60d_avg > 0 (right) compared to prediction probability percentile bucket.

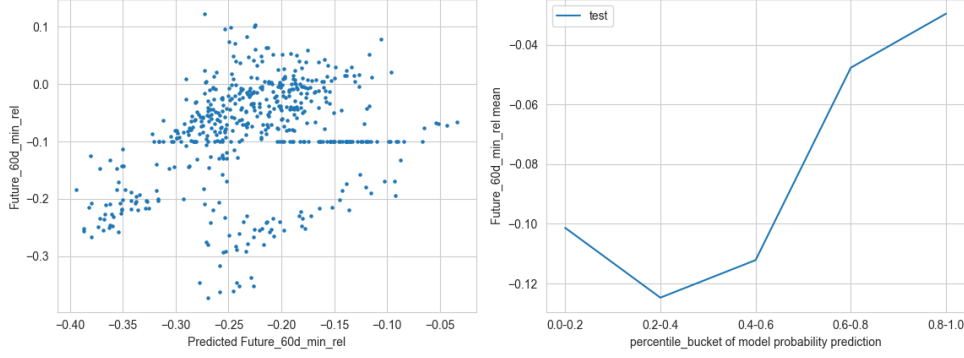


Figure 9. *Future_60d_min_rel* compared to XGBoost test set prediction (left) [*corr*: 0.31, *p-val*: $5e-13$] and compared to prediction percentile bucket (right).

3.2. Walk-forward prediction

To generate the predictions for our backtesting window (2018-01-01 to 2025-12-31), the model is trained with walk-forward prediction. Each `CYCLE_LEN`-day period is predicted using all prior data up to `PURGE_LEN` days before the start of the current cycle. The `PURGE_LEN` data exclusion is intended to avoid forward data leakage from the target metric, e.g., `target_up_60d` will contain information up to the following 60 days profitability.

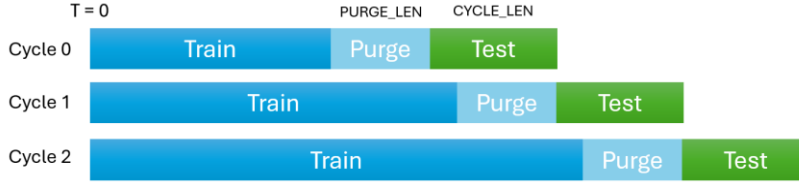


Figure 10. Illustration of walk forward prediction with purge period to avoid forward leakage.

Table 7. Selected `CYCLE_LEN` and `PURGE_LEN` for different target metrics

Target metric	<code>target_up_60d</code>	<code>ROI_30d_top30</code>	<code>Future_60d_min_rel</code>
<code>CYCLE_LEN</code>	5	5	90
<code>PURGE_LEN</code>	60	30	60

The walk-forward prediction for `target_up_60d` and `Future_60d_min_rel` exhibited poor correlation to target metrics (Figure 11, Figure 13), in contrast to the simple train-test split. The model performance significantly degrades during walk-forward validation. This indicates the signals identified during static validation do not generalize under realistic, sequential retraining situations.

On the other hand, training XGBoost on `ROI_30d_top30` to produce prediction probability `pred_top30` maintained its correlation to `ROI_60d_avg` although a lot of noise was also observed. Monotonous increase in `ROI_60d_avg` and fraction of `ROI_60d_avg > 0` was also observed when data was broken down by quintile buckets (Figure 14), suggesting signal is consistent. One probable reason this target metric is more robust than `target_up_60d` is the larger separation from zero, which forced the model to learn only high confidence signals.

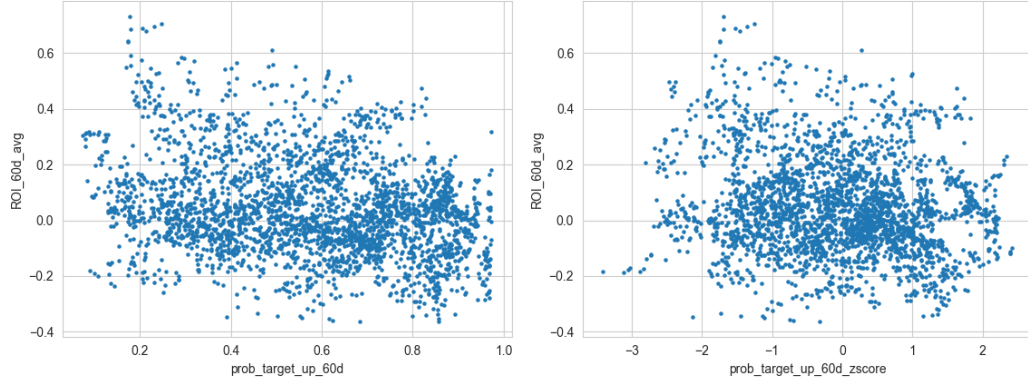


Figure 11. Walk-forward prediction probability *prob_target_up_60d* (left) and *prob_target_up_60d_zscore* (right) compared to *ROI_60d_avg*. No clear correlation found.

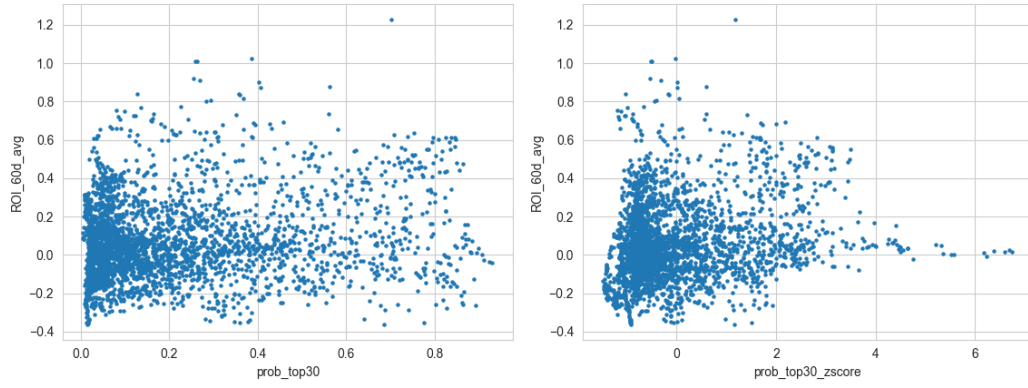


Figure 12. Walk-forward prediction probability *prob_top30* (left) and *prob_top30_zscore* (right) compared to *ROI_60d_avg*. A slight correlation observed.

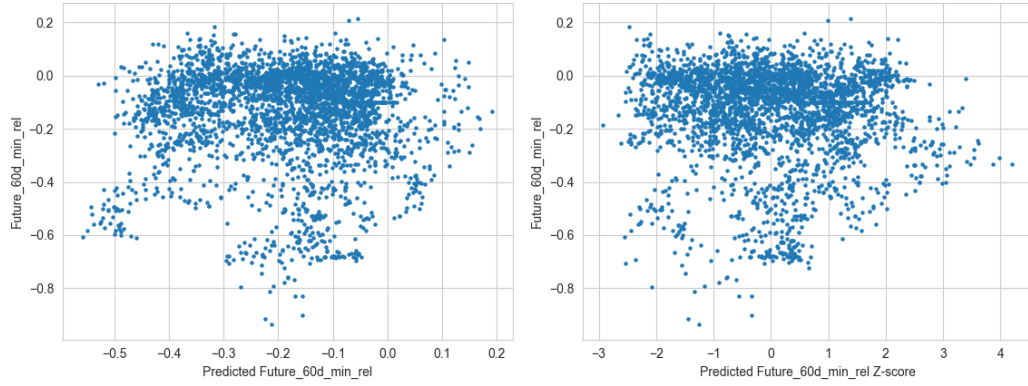


Figure 13. Walk-forward of *Future_60d_min_rel* (left) and the *z-score* (right) compared to actual values. Correlation is observed to be weak.

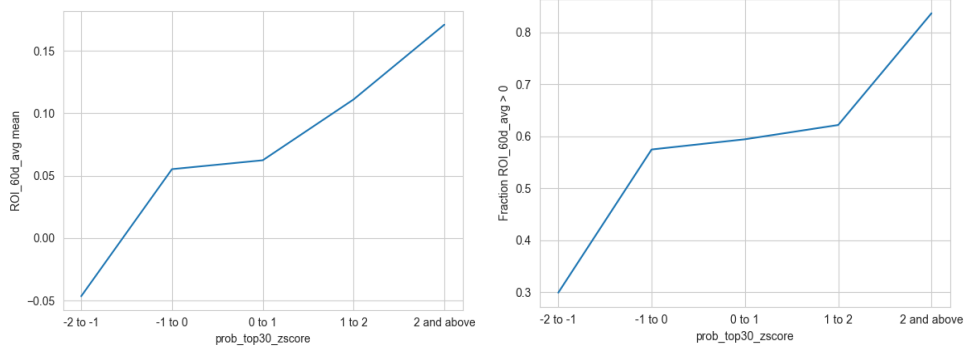


Figure 14. ROI_60d_avg mean (left) and fraction of ROI_60d_avg > 0 (right) compared to prob_top30_zscore quintiles. Monotonic increase observed in both plots.

3.3. Feature importance

The SHAP (SHapley Additive exPlanations) values uses a game-theoretic approach to quantify the contribution of each feature to the predicted output of machine learning models [7]. The SHAP values from ROI_30d_top30 are reviewed here as this model showed the strongest signal.

The top 10 features and the directional impact of each feature are shown in Figure 15. Each instance is placed on individual feature row, and the x-axis represents the impact to model output. For example, low (blue points) from “distance_sma_200” feature have positive model output impact. The model probability changes with each instance (time) and with varying feature contributions as seen in the heatmap Figure 16.

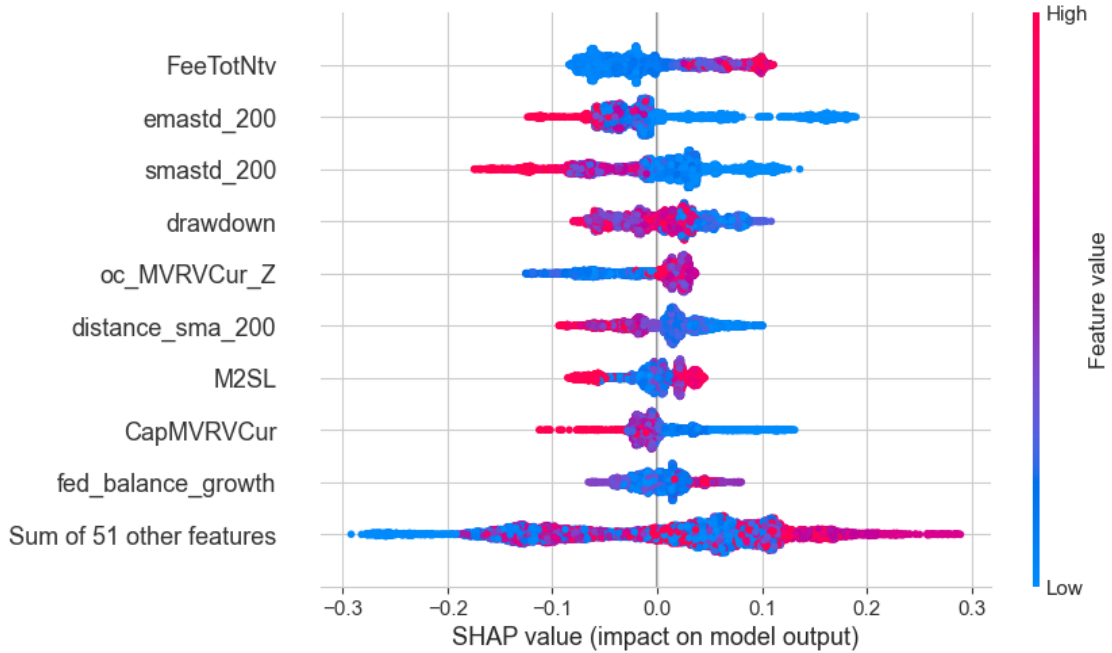


Figure 15. SHAP beeswarm plot showing the top 10 most significant features and directional impact.

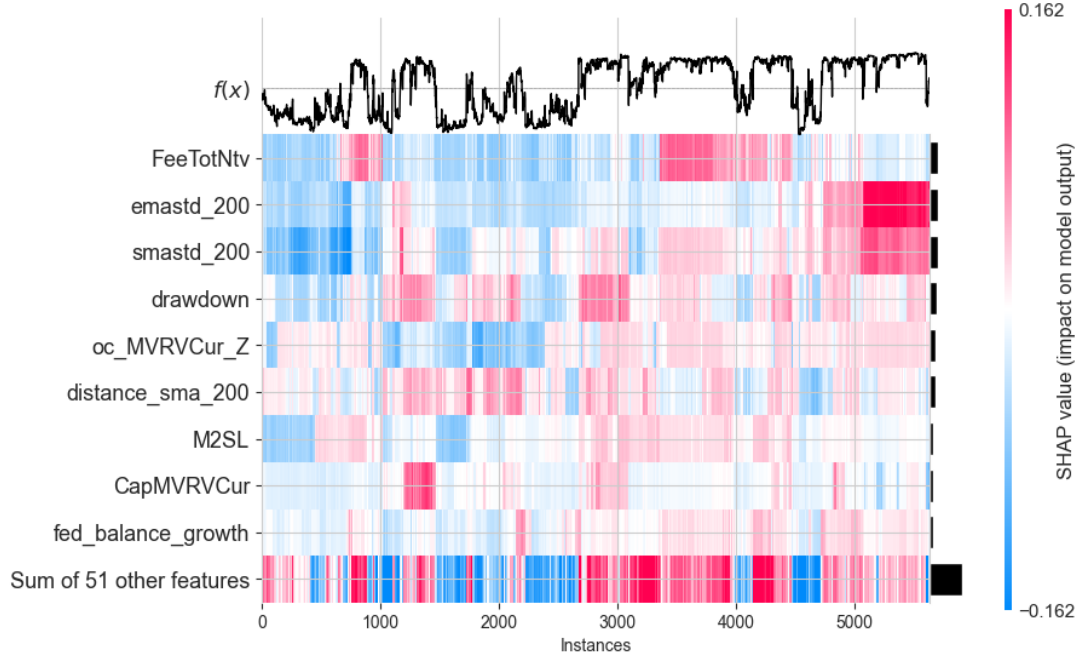


Figure 16. SHAP heatmap plot shows final model probability and contribution from top features.

Table 8 lists directional impact of the top features and the possible explanations why these relationships are observed. The relationships are intuitive except for `oc_MVRVCur_Z` (MVRV Z-score) which suggests higher returns are more likely when the Z-score is high. This means the model captured the upwards momentum when Bitcoin prices are increasing relative to the Z-score rolling window, which is 365 days for this metric.

Table 8. Top 10 features, directional impact, and probable explanation

Feature name	Positive impact to model output when feature value is	Possible explanation
FeeTotNtv	High	High network usage
emastd_200	Low	Low volatility is less risky for accumulation
smastd_200	Low	Low volatility is less risky for accumulation
drawdown	Low	Buy when cheaper
oc_MVRVCur_Z	High	Opposite of expectation, captures momentum instead of value signal
distance_sma_200	Low	Buy when cheaper
CapMVRVCur	Low	Value signal to buy when low
fed_balance_growth	High	Indicates quantitative easing and more cash injection

Figure 17 (left) shows the impact of the top features which exhibit distributed contribution instead of a small number of dominant features. Figure 17 (right) categorizes each feature and shows that Technical, On-Chain, and Macro features dominated while Polymarket and Binance features had limited contribution. Hence, despite extensive work on Polymarket features, the model was unable to

utilize it. One explanation is the complete Polymarket data is only available since April 2025, which limits its testing to very recent time periods. With more data availability in the future, Polymarket data may prove to be useful for Bitcoin accumulation signal.

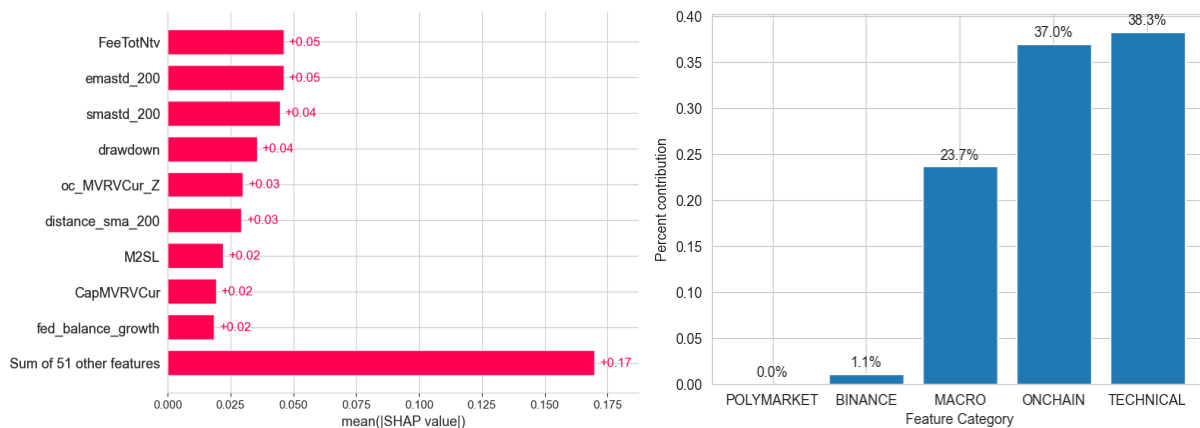


Figure 17. Feature contribution magnitude (left) and percentage contribution by feature type (right).

4. Strategy Development

4.1. Setup

Below are my considerations for the design of the DCA strategy combining heuristics and model-driven signals to balance valuation, momentum, and risk:

- Buy more units when price is low. This will prevent us from buying at peak prices. Included features: MVRV Z-score (Figure 18), normalized price compared to moving average (price_vs_ma), and relative strength index (RSI) indicating whether an asset is oversold or overbought.

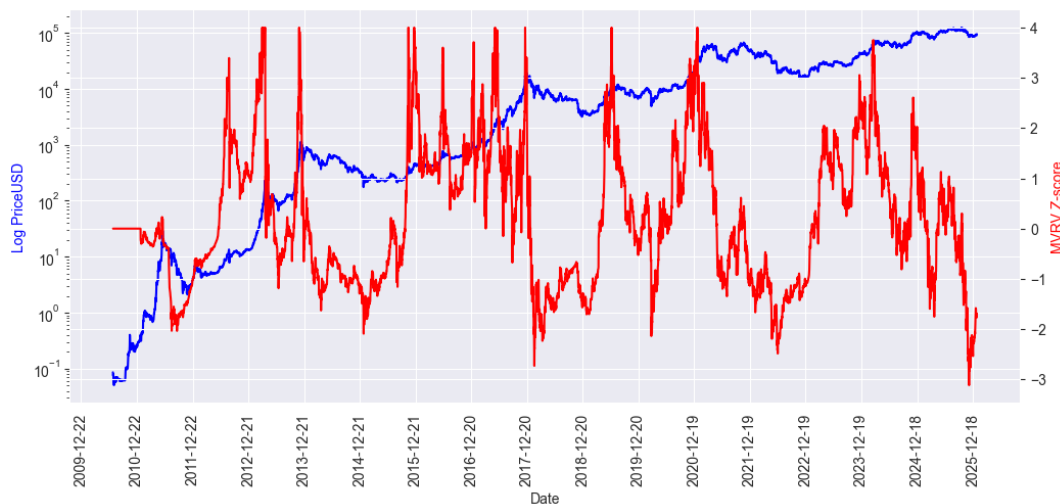
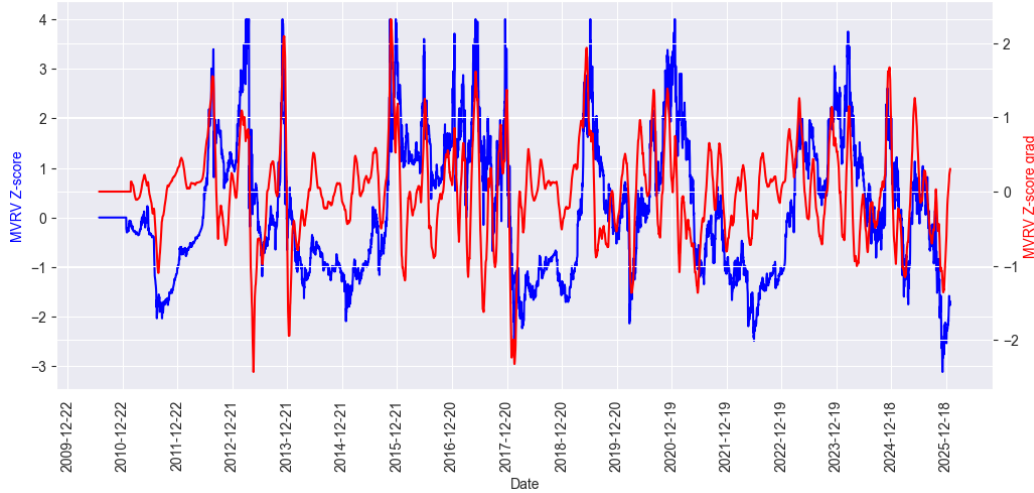


Figure 18. Bitcoin log PriceUSD and MVRV Z-score

- Price change today is correlated to the next (1st derivative or gradient). Allocation can be increased during upward momentum regimes and lowered when prices are decreasing. Included features: MVRV Z-score gradient (1st derivative).



- Look out for trend reversals (2nd derivative or acceleration). Price increases will slow down when overbought and will increase selling pressure. Included features: MVRV Z-score acceleration (2nd derivative).
- Price volatility is a signal for risk as price changes become less predictable. Included features: log price volatility Z-score, MVRV volatility Z-score.
- XGBoost generated predictions will also be included. Between the two profitability-oriented models, the one based on ROI_30d_top30 metric is better performing and will be included in ml1_score. The risk-oriented metric from Future_60d_min_rel will also be used for ml2_score.

I decided to combine the various signals by multiplying them together, setting 1 as the default or neutral score:

$$\text{multiplier} = \text{valuation_score} \times \text{grad_score} \times \text{accel_score} \times \text{ml1_score} \times \text{ml2_score} \times \text{risk_score}$$

- valuation_score: combine three different valuation signals and get the exponential. This means positive signal will have exponentially larger signal.

$$v = v_1(\text{MVRV Z-score}) + v_2(\text{price_vs_ma}) + v_3(\text{RSI Z-score}). v_1 + v_2 + v_3 = 1$$

$$\text{valuation_score} = \exp(\text{clip}(v, -v_{lim}, v_{lim}))$$

- grad_score: regime dependent, adjustment is linear to the MVRV Z-score gradient.

$$g = \begin{cases} 1 + g_1 \min(\text{MVRV Z-score grad}, 1), & \text{MVRV Z-score} > 1.5 \\ 1, & \text{Otherwise} \\ 1 + g_2 \max(\text{MVRV Z-score grad}, -1), & \text{MVRV Z-score} < -1 \end{cases}$$

$$grad_score = \exp(\text{clip}(g, g_{min}, g_{max}))$$

- accel_score: increase weightage if gradient and acceleration have the same sign, vice versa.

$$a = \begin{cases} 1 + a_1|MVRV \text{ Z-score accel}|, & \text{accel and grad same sign} \\ 1 - a_2|MVRV \text{ Z-score accel}|, & \text{accel and grad opposite sign} \end{cases}$$

$$accel_score = \exp(\text{clip}(a, a_{min}, a_{max}))$$

- ml1_score: increase multiplier if prob_top30 Z-score is above a certain threshold.

$$ml1_score = 1 + \frac{m_1}{1.5} \text{clip}(\text{prob_top30_Z-score}, 0.5, 2)$$

- ml2_score: decrease multiplier if pred_future_60d_min_rel is below a certain threshold.

$$ml2_score = 1 - \frac{m_2}{0.2} \text{clip}(\text{pred_future_60d_min_rel}, -0.4, -0.2)$$

- risk_score: reduces allocation if high price or MVRV volatility.

$$r = -r_1(\log\text{price volatility Z-score}) - r_2(\text{MVRV volatility Z-score})$$

$$risk_score = 1 - \frac{r_3}{2} \text{clip}(r, -3, -1)$$

The strategy was implemented with initial best guess parameters. To explore the parameter space, an optimization framework from Optuna library was implemented for 150 cycles.

4.2. Results and analysis

Performance is evaluated using satoshi-per-dollar (SPD), which measures Bitcoin accumulated per unit capital deployed, and win-rate against Uniform DCA in 1-year rolling windows. The final model with best-guess parameters achieved model score of 58.2% based on combined 54.76% SPD exp-decay average and 61.63% win-rate compared to uniform DCA. The model with optimized parameters further improved the score to 60.16%, beating other models in the comparison table. The optimized model further improved the SPD exp-decay average to 57.63% and the win-rate to 62.69% and achieved approximately 14,000% cumulative performance advantage relative to Uniform DCA.

Table 10 shows compared to the best guess parameters, the optimized model overweighted valuation_score and MVRV Z-score. On the other hand, the optimal model underweighted RSI, accel_score, ml2_score, and risk_score. The results indicate that despite extensive efforts to generate more features, many were not beneficial and only added noise to the strategy.

Table 9. Metrics summary for different models

Metric	Price_vs_ma	Example 1	Final (best-guess parameters)	Final (optimized parameters)
Model score	51.74%	59.54%	58.2%	60.16%
Exp-decay avg	42.43%	58.78%	54.76%	57.63%
Win rate	61.05%	60.31%	61.63%	62.69%
Mean excess	1.04%	5.7%	3.09%	5.41%
Median excess	1.23%	6.43%	5.24%	8.23%
Relative improvement	2.61%	16.82%	9.19%	15.28%
Ratio	1.03	1.17	1.09	1.15

Table 10. Selected differences between best-guess and optimal model parameters

Weights	Remark	Final (best-guess)	Final (optimized)
v_1	MVRV Z-score weight	0.5	0.63
v_3	RSI weight	0.15	0.007
v_{lim}	Valuation score range	3	4.9
a_1	Acceleration multiplier for same direction	0.3	0.007
a_2	Acceleration multiplier for opposite direction	0.2	0.01
m_2	ML2 multiplier	0.3	0.036
r_3	Risk score multiplier	0.2	0.056

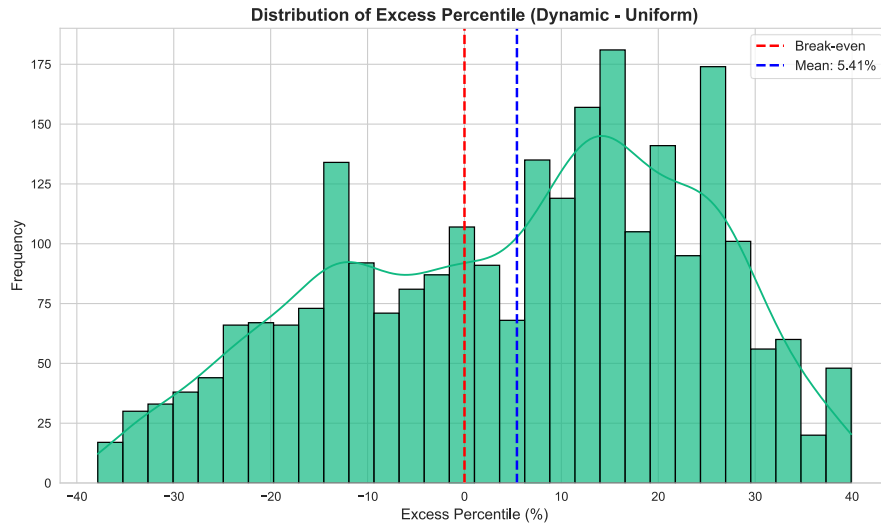


Figure 19. Distribution of excess SPD percentile for Dynamic DCA compared to Uniform DCA

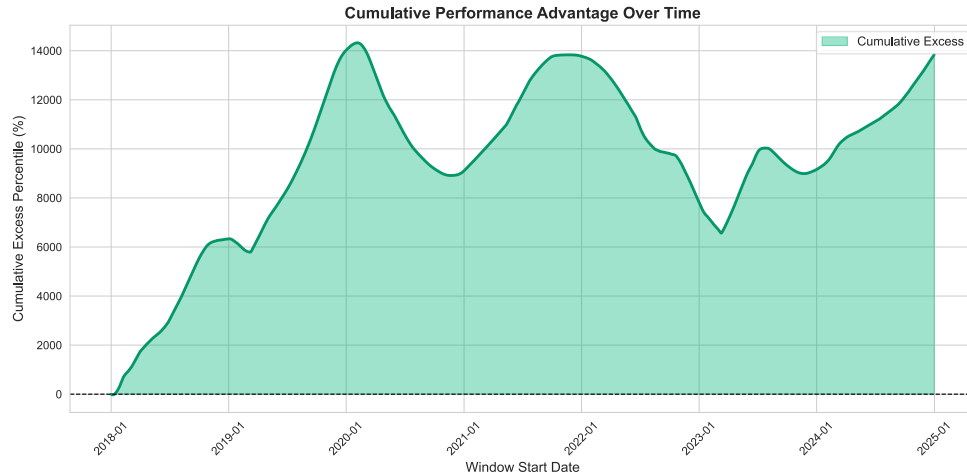


Figure 20. Cumulative performance advantage over time

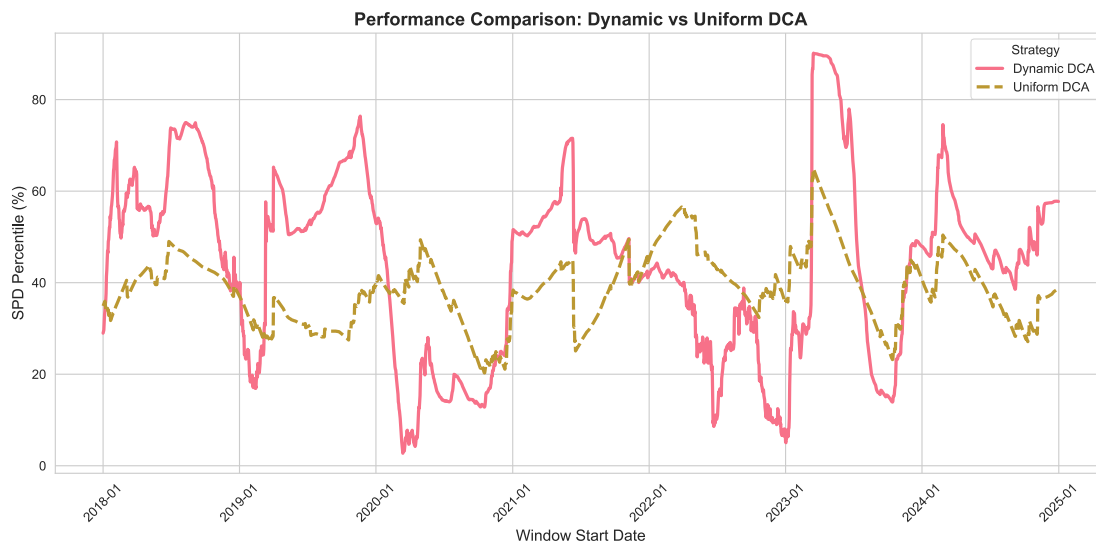


Figure 21. Performance comparison between final model and uniform DCA

Figure 21 shows the final model underperformed to Uniform DCA for backtesting in two significant periods:

- In 2020, there was a stratospheric price increase which caused the model to reduce Bitcoin accumulation. However, the strategy failed to predict that price would continue to increase significantly. This reflects the model's inability to extrapolate beyond previously observed regimes. One potential improvement is to increase the range of the grad_score component to capture more of the upside.
- On the other hand, the underperformance in 2022 is because the model continued to accumulate Bitcoin as the price was dropping to below 2021 levels. We can potentially improve the model by extending the MVRV Z-score window to include the pre-2021 price levels.

5. Conclusions

Extensive feature engineering was performed across on-chain, technical, macroeconomic, and prediction market datasets.

Among the evaluated targets, ROI_30d_top30 produced most robust and consistent predictive signal under walk-forward validation. Despite extensive feature engineering, Polymarket data was not utilized by the model, possibly because the data is only available for limited date range.

The final Dynamic DCA strategy combined valuation, gradient, acceleration, ML1 (prediction probability for ROI_30d_top30), ML2 (predicted Future_60d_min_rel), and risk scores. Model parameter optimization further improved the final score to 60.16% and significantly outperformed Uniform DCA.

Future model improvements should focus on improving robustness across regime shifts and generalization under walk-forward retraining conditions.

6. References

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